ADULT HYPOKALAEMIA ALGORITHM

GUIDELINES 2025
EUROPEAN RESUSCITATION COUNCIL©

- Assess using ABCDE approach
- 12-lead ECG and monitor cardiac rhythm if serum potassium (K^+) < 3.0 mmol L^{-1}
- Increased risk in patients taking digoxin
- Remember hypokalaemia is exacerbated by magnesium deficiency (< 0.6 mmol L^{-1})

MODERATE
K^+ 2.5 - 2.9 mmol L^{-1}

SEVERE
K^+ ≤ 2.4 mmol L^{-1}

CARDIAC ARREST

Symptomatic or unable to take oral replacement

Emergency treatment indicated

- **ECG changes -** U waves, prolonged PR interval, T wave flattening / inversion, ST depression, arrhythmias (increased risk if taking Digoxin)
- **Effect on muscle -** paralysis, rhabdomyolysis, respiratory difficulties

- **Start advanced life support**
- **Give adrenaline**
- **Confirm hypokalaemia**

Rate guided by clinical urgency

IV potassium
(40 mmol KCl in 1000 ml 0.9% NaCl)

Standard rate: 10 mmol hr^{-1}

Maximum rate: 20 mmol hr^{-1} (in HDU / ICU)

Peri-arrest: 20 mmol over 10 min, followed by 10 mmol over 10 min. Start infusion and tailor rate

Rate guided by clinical urgency

IV magnesium (if required)
50% magnesium sulphate (1 g = 4 mmol)

Standard rate: 2 g (8 mmol) over 4 hrs

Medical emergency: 2 g (8 mmol over 10-15 min (may be repeated once if necessary)

IV potassium

Cardiac arrest: 20 mmol over 2-3 min, followed by 10 mmol over 2 min

Start infusion until serum K^+ ≥ 4.0 mmol L^{-1}

IV magnesium (if required)
50% magnesium sulphate (1 g = 4 mmol)

Cardiac arrest: 2 g (8 mmol) over 2 min

Monitor serum K^+ and blood glucose

(Adjust rate of infusion according to K^+ level; target K^+ level = 4.0 mmol L^{-1})

Seek expert help

ABCDE – airway, breathing, circulation, disability, exposure; ECG – electrocardiogram; IV – intravenous; HDU - high dependency unit; ICU - intensive care unit; K - potassium; Mg - magnesium